



Machine Learning Methodology for E-commerce Orders

(Classification: Predict Order Status)



Target Variable: **Status**

1

Dataset Selection



- Dataset: clean_final_data.csv
- 49,222 rows, 17 columns
- E-commerce orders data (customer, product, payment, and sales information)

2

Data Understanding



- Explore data shape and columns
- Check data types and missing values
- Understand features and target variable (Status)
- Analyze distributions and class balance

3

Data Preprocessing



- Handle missing values (if any)
- Convert date columns (OrderDate, SignupDate)
- Encode categorical variables (e.g., PaymentMethod, City)
- Feature scaling (if needed)
- Prepare data for modeling

4

Train/Test Split



- Split the data into training and testing sets
- Training set: 80%
- Testing set: 20%
- Use stratified split to maintain the same class distribution (Completed, Cancelled, Returned)

5

Model Training



- Use classification algorithms (e.g., Logistic Regression, Decision Tree, Random Forest)
- Train the model on the training data
- Tune hyperparameters (if needed)

6

Model Evaluation



- Evaluate the model on test data
- Metrics: Accuracy, Precision, Recall, F1-Score
- Use confusion matrix
- Compare different models to select the best one

7

Results & Insights



- Predict order status: Completed, Cancelled, or Returned
- Identify important features that influence the prediction
- Provide business insights to improve customer experience
- Finalize and save the best model

Order Status Classes

● Completed

● Cancelled

● Returned

Turning Data into
Better Customer Experiences

